Muhammad Arslan, Hussam Ghanem, Saba Munawar, and Christophe Cruz. A survey on rag with llms. *Procedia Computer Science*, 246:3781–3790, 2024.

Marthe Ballon, Andres Algaba, and Vincent Ginis. The relationship between reasoning and performance in large language models–o3 (mini) thinks harder, not longer. *arXiv preprint arXiv:2502.15631*, 2025.

Yoshua Bengio, Michael Cohen, Damiano Fornasiere, Joumana Ghosn, Pietro Greiner, Matt MacDermott, Sören Mindermann, Adam Oberman, Jesse Richardson, Oliver Richardson, et al. Superintelligent agents pose catastrophic risks: Can scientist ai offer a safer path? *arXiv preprint arXiv:2502.15657*, 2025.

Xiaohe Bo, Zeyu Zhang, Quanyu Dai, Xueyang Feng, Lei Wang, Rui Li, Xu Chen, and Ji-Rong Wen. Reflective multi-agent collaboration based on large language models. *Advances in Neural Information Processing Systems*, 37:138595–138631, 2024.

Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.

Garyk Brixi, Matthew G Durrant, Jerome Ku, Michael Poli, Greg Brockman, Daniel Chang, Gabriel A Gonzalez, Samuel H King, David B Li, Aditi T Merchant, et al. Genome modeling and design across all domains of life with evo 2. *BioRxiv*, pages 2025–02, 2025.

Yuji Cao, Huan Zhao, Yuheng Cheng, Ting Shu, Yue Chen, Guolong Liu, Gaoqi Liang, Junhua Zhao, Jinyue Yan, and Yun Li. Survey on large language model-enhanced reinforcement learning: Concept, taxonomy, and methods. *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

Mert Cemri, Melissa Z Pan, Shuyi Yang, Lakshya A Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, et al. Why do multi-agent llm systems fail? *arXiv preprint arXiv:2503.13657*, 2025.

Miaosen Chai, Emily Herron, Erick Cervantes, and Tirthankar Ghosal. Exploring scientific hypothesis generation with mamba. In *Proceedings of the 1st Workshop on NLP for Science (NLP4Science)*, pages 197–207, 2024.

Jun Shern Chan, Neil Chowdhury, Oliver Jaffe, James Aung, Dane Sherburn, Evan Mays, Giulio Starace, Kevin Liu, Leon Maksin, Tejal Patwardhan, et al. Mle-bench: Evaluating machine learning agents on machine learning engineering. *arXiv preprint arXiv:2410.07095*, 2024.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, Wei Ye, Yue Zhang, Yi Chang, Philip S. Yu, Qiang Yang, and Xing Xie. A survey on evaluation of large language models. *ACM Trans. Intell. Syst. Technol.*, 15(3), March 2024. ISSN 2157-6904. doi: 10.1145/3641289. URL https://doi.org/10.1145/3641289.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde De Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, Botao Yu, Yifei Li, Zeyi Liao, Chen Wei, Zitong Lu, Vishal Dey, Mingyi Xue, Frazier N. Baker, Benjamin Burns, Daniel Adu-Ampratwum, Xuhui Huang, Xia Ning, Song Gao, Yu Su, and Huan Sun. Scienceagentbench: Toward rigorous assessment of

language agents for data-driven scientific discovery. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=6z4YKr0GK6.

Corinna Cortes and Neil D Lawrence. Inconsistency in conference peer review: Revisiting the 2014 neurips experiment. *arXiv preprint arXiv:2109.09774*, 2021.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. A survey on rag meeting llms: Towards retrieval-augmented large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 6491–6501, 2024.

Aniketh Garikaparthi, Manasi Patwardhan, Lovekesh Vig, and Arman Cohan. Iris: Interactive research ideation system for accelerating scientific discovery. *arXiv preprint arXiv:2504.16728*, 2025.

Alireza Ghafarollahi and Markus J Buehler. Sciagents: Automating scientific discovery through multi-agent intelligent graph reasoning. *arXiv preprint arXiv:2409.05556*, 2024.

Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Anil Palepu, Petar Sirkovic, Artiom Myaskovsky, Felix Weissenberger, Keran Rong, Ryutaro Tanno, et al. Towards an ai co-scientist. *arXiv preprint arXiv:2502.18864*, 2025.

Tianyang Gu, Jingjin Wang, Zhihao Zhang, and HaoHong Li. Llms can realize combinatorial creativity: generating creative ideas via llms for scientific research. *arXiv preprint arXiv:2412.14141*, 2024a.

Yang Gu, Hengyu You, Jian Cao, Muran Yu, Haoran Fan, and Shiyou Qian. Large language models for constructing and optimizing machine learning workflows: A survey. *arXiv preprint arXiv:2411.10478*, 2024b.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xiangliang Zhang. Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680*, 2024.

Yichen He, Guanhua Huang, Peiyuan Feng, Yuan Lin, Yuchen Zhang, Hang Li, et al. Pasa: An llm agent for comprehensive academic paper search. *arXiv preprint arXiv:2501.10120*, 2025.

Xinyi Hou, Yanjie Zhao, Shenao Wang, and Haoyu Wang. Model context protocol (mcp): Landscape, security threats, and future research directions. *arXiv preprint arXiv:2503.23278*, 2025.

Xiang Hu, Hongyu Fu, Jinge Wang, Yifeng Wang, Zhikun Li, Renjun Xu, Yu Lu, Yaochu Jin, Lili Pan, and Zhenzhong Lan. Nova: An iterative planning and search approach to enhance novelty and diversity of llm generated ideas. *arXiv preprint arXiv:2410.14255*, 2024.

Changwu Huang, Zeqi Zhang, Bifei Mao, and Xin Yao. An overview of artificial intelligence ethics. *IEEE Transactions on Artificial Intelligence*, 4(4):799–819, 2022.

Intology. Zochi technical report. *arXiv*, 2025.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*, 2024.

Peter Jansen, Oyvind Tafjord, Marissa Radensky, Pao Siangliulue, Tom Hope, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Daniel S Weld, and Peter Clark. Codescientist: End-to-end semi-automated scientific discovery with code-based experimentation. *arXiv preprint arXiv:2503.22708*, 2025.

Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. Aide: Ai-driven exploration in the space of code. *arXiv preprint arXiv:2502.13138*, 2025.

Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. Swe-bench: Can language models resolve real-world github issues? In *ICLR*, 2024.

Anna Jobin, Marcello Ienca, and Effy Vayena. The global landscape of ai ethics guidelines. *Nature machine intelligence*, 1(9):389–399, 2019.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *nature*, 596(7873):583–589, 2021.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Arif Ali Khan, Sher Badshah, Peng Liang, Muhammad Waseem, Bilal Khan, Aakash Ahmad, Mahdi Fahmideh, Mahmood Niazi, and Muhammad Azeem Akbar. Ethics of ai: A systematic literature review of principles and challenges. In *Proceedings of the 26th international conference on evaluation and assessment in software engineering*, pages 383–392, 2022.

Ross D King, Jem Rowland, Stephen G Oliver, Michael Young, Wayne Aubrey, Emma Byrne, Maria Liakata, Magdalena Markham, Pinar Pir, Larisa N Soldatova, et al. The automation of science. *Science*, 324(5923): 85–89, 2009.

Patrick Tser Jern Kon, Jiachen Liu, Qiuyi Ding, Yiming Qiu, Zhenning Yang, Yibo Huang, Jayanth Srinivasa, Myungjin Lee, Mosharaf Chowdhury, and Ang Chen. Curie: Toward rigorous and automated scientific experimentation with ai agents. *arXiv preprint arXiv:2502.16069*, 2025.

Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. Llms get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*, 2025.

P Langley. *Scientific discovery: Computational explorations of the creative processes*. MIT Press, 1987.

Long Li, Weiwen Xu, Jiayan Guo, Ruochen Zhao, Xingxuan Li, Yuqian Yuan, Boqiang Zhang, Yuming Jiang, Yifei Xin, Ronghao Dang, et al. Chain of ideas: Revolutionizing research via novel idea development with llm agents. *arXiv preprint arXiv:2410.13185*, 2024a.

Ruochen Li, Liqiang Jing, Chi Han, Jiawei Zhou, and Xinya Du. Learning to generate research idea with dynamic control. *arXiv preprint arXiv:2412.14626*, 2024b.